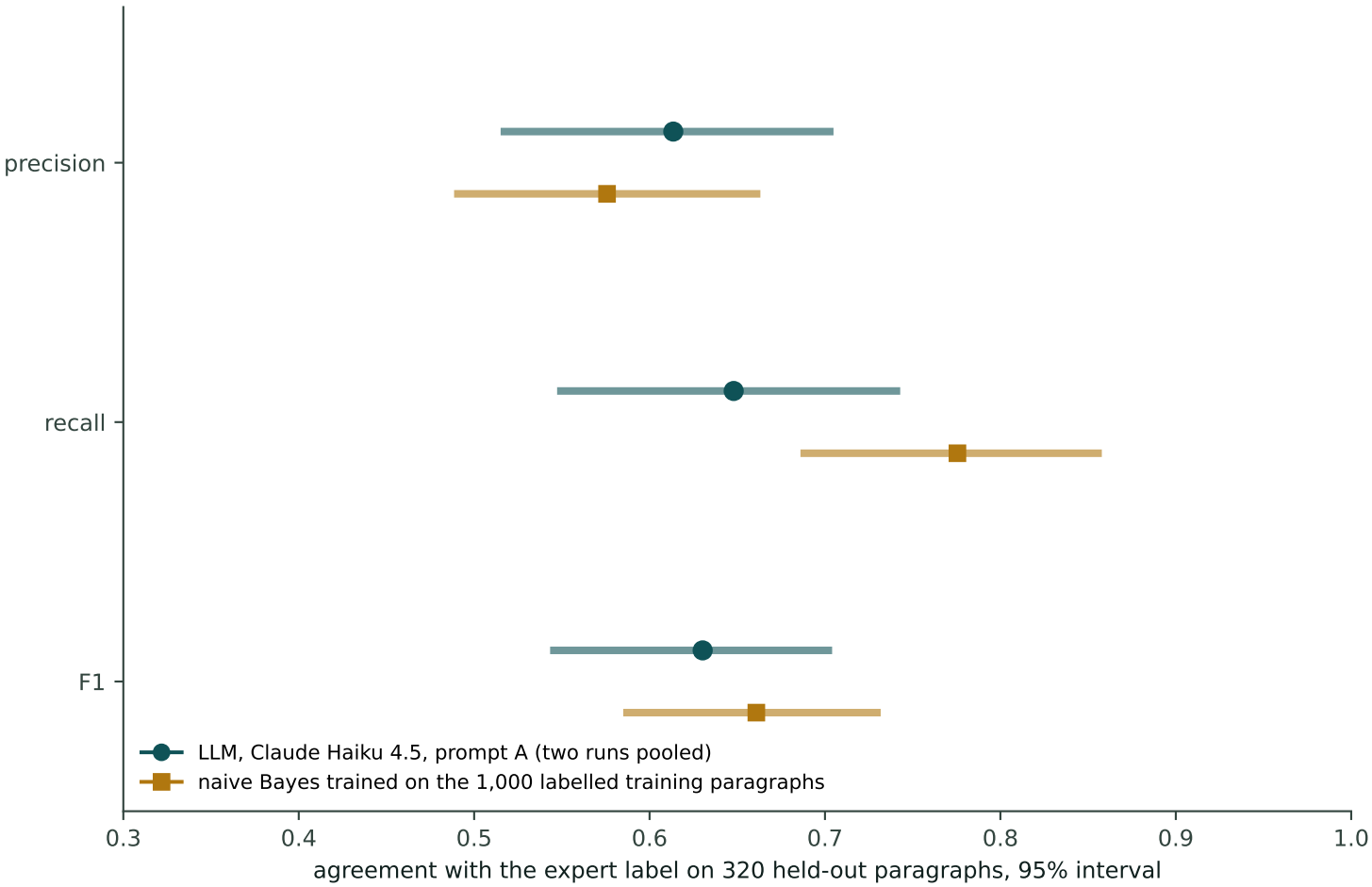
How right is an LLM that reads climate disclosures?

The estimand

Precision and recall of an LLM's yes/no extraction of climate commitments and actions, against the expert label, on paragraphs from corporate annual and sustainability reports. If I knew it exactly it would be the share of the model's 'commitment' calls that experts also call so, and the share of expert- labelled commitments it finds, for paragraphs drawn like these. The reference labels are the public ClimateBERT annotations, not my own hand coding.

The answer

Precision 61.4% [51.7%, 70.3%], recall 64.8% [54.9%, 74.1%], F1 63.0% [54.5%, 70.2%]. Kappa against the experts is 0.47 and 0.45 on the two runs: moderate agreement. About 39 in 100 calls it makes are not commitments to the experts, and it misses 35 in 100 that are. A naive Bayes trained on 1,000 labelled paragraphs scores F1 66.1%: no worse, and no better.



Method: held-out labels, fixed prompt, resampling

Data

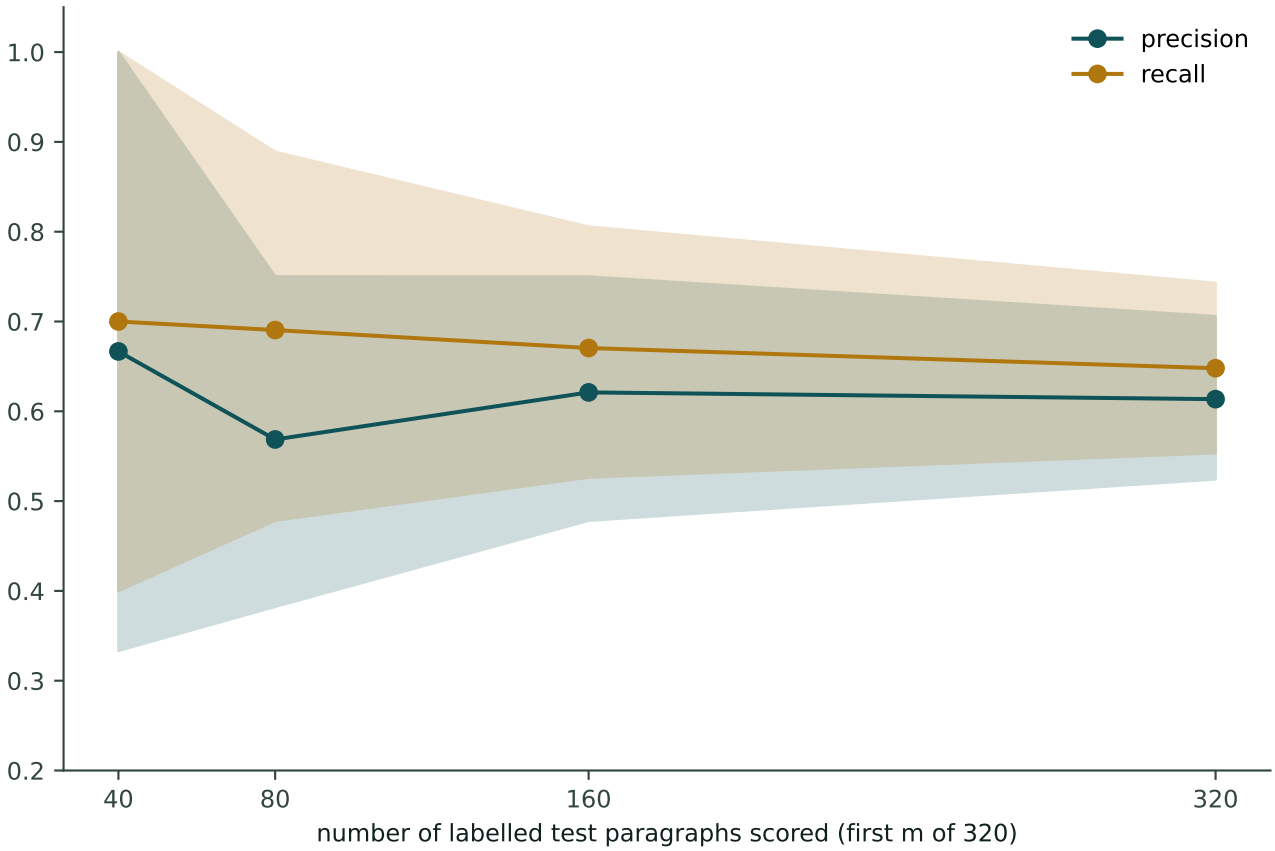
ClimateBERT 'commitments and actions' dataset (climatebert/climate_commitments_actions), fetched from the Hugging Face datasets-server on 2026-10-26: 1,000 training and 320 test paragraphs, each labelled by experts as about climate commitments and actions or not. Test prevalence 30.6%, training 42.5%. CC-BY-NC-SA text is fetched, not committed.

Extraction

Claude Haiku 4.5 via the Claude Code CLI, tools and settings off, one paragraph per call, 1,600 calls in all. Primary: prompt A, the dataset card's own task definition, written before any reply was read. Run twice. Sweep: prompt B (terse), prompt C (strict: target, pledge or concrete action), and Claude Sonnet 5.5 on A.

Interval

4,000 resamples of the 320 test paragraphs, the population the estimate belongs to. In each resample every paragraph takes one of the two runs at random, so the model's own sampling noise is inside the interval. The prompt is one choice for all paragraphs, so it is swept, not drawn. Labelling error is not in the interval.



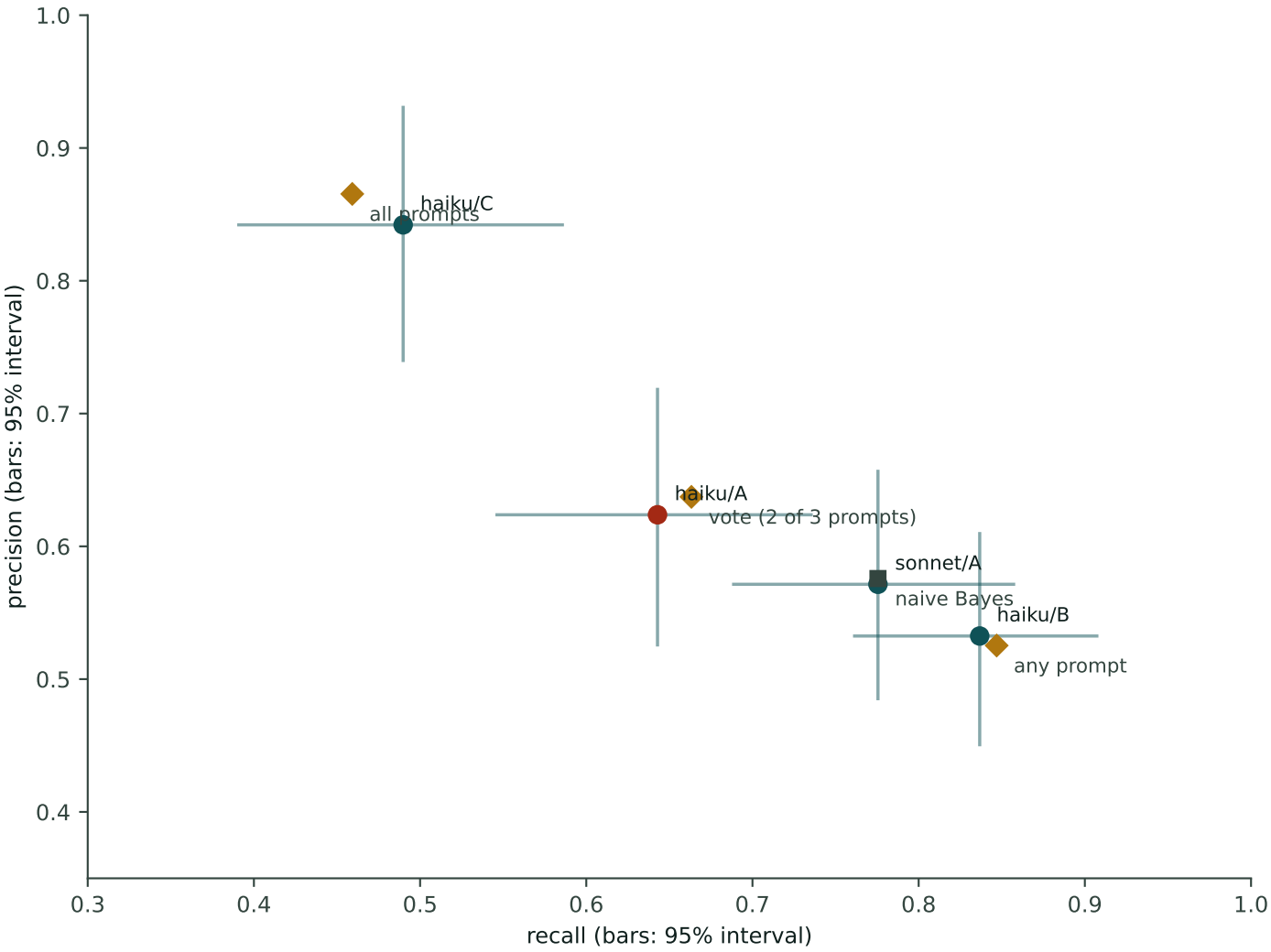
Result: the prompt moves the trade-off, not F1

What moves

Across 4 model and prompt combinations and three ways of combining prompts, precision runs from 52.5% to 86.5% and recall from 45.9% to 84.7%, while F1 stays between 60.0% and 65.8%. The strict prompt C is the most precise (84.2%, recall 49.0%); the terse prompt B finds the most (83.7%, precision 53.2%). Sonnet 5.5 on prompt A: F1 65.8% [58.2%, 72.6%] against 63.3% [55.1%, 70.6%] for Haiku: not separable.

What it means

Which prompt to use is a choice about the cost of a miss against the cost of a false alarm, not a search for the better model. A screening use wants B's recall; an indicator that is published wants C's precision.



Validation and sensitivity

Against a model I trained, on the same paragraphs

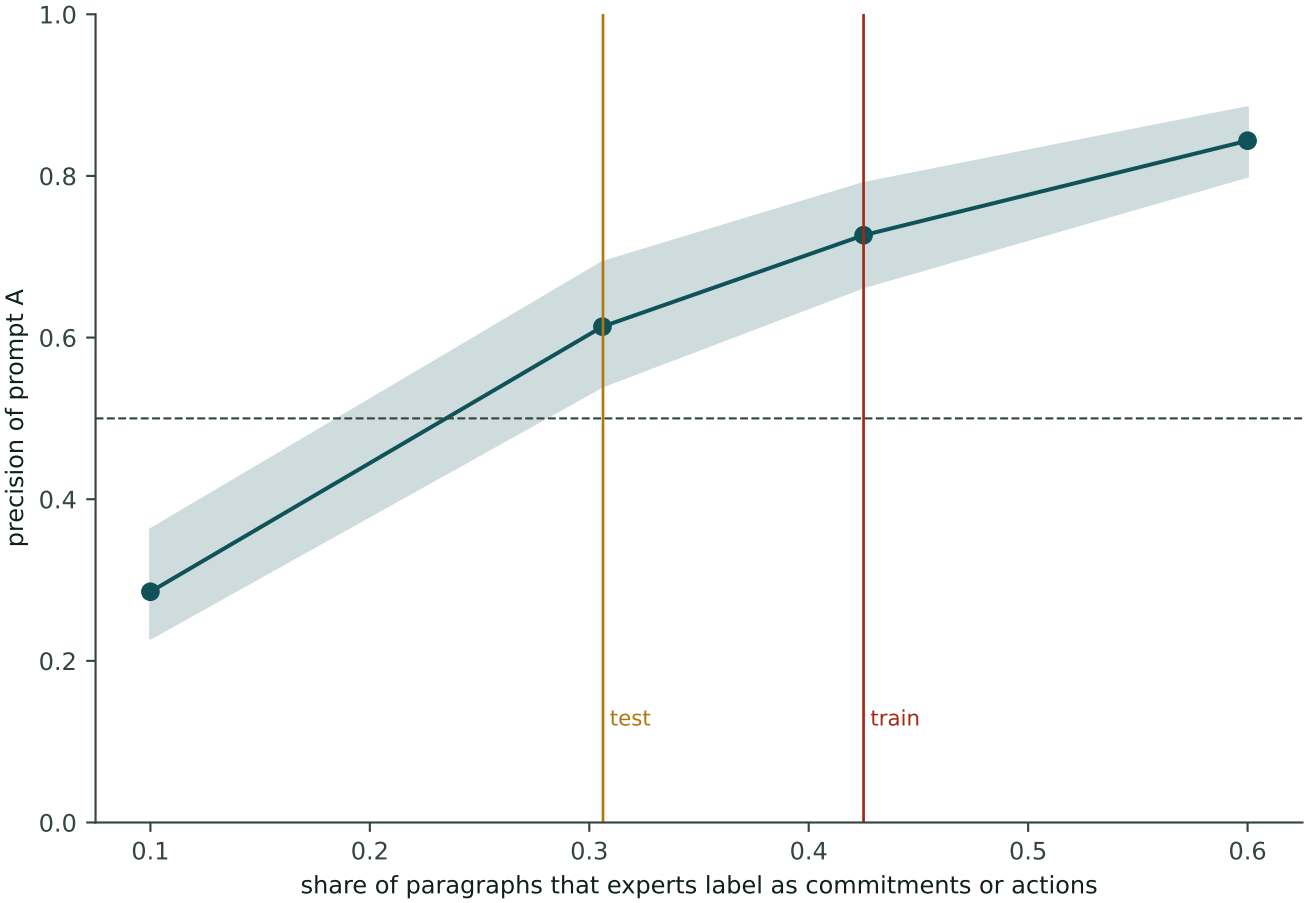
LLM minus naive Bayes, paired resamples: precision +3.8 points [-4.9, +12.9], recall -12.6 [-24.1, -0.9], F1 -3.0 [-11.4, +5.8]. The zero-shot model has no precision edge, and it finds fewer of the commitments.

Test-retest, same model, same prompt, same paragraphs

The two runs disagree on 19 of 320 paragraphs (5.9%); kappa between them 0.86. The model is far more consistent with itself than with the experts (kappa 0.47): most of the error is systematic, not noise.

Precision depends on the mix, recall does not

Precision is 28.5% where 10.0% of paragraphs are commitments, 61.4% at the test mix and 72.7% at the training mix. A document with few commitments gets mostly false alarms. Sample size: the precision interval is 37 points wide at 80 paragraphs and 18 at 320 (page 2). Length: F1 by tercile 62.0%, 67.6%, 59.0%; the three intervals overlap, so no length effect is claimed.



Limits and what comes next

The assumption that would break the conclusion

That the expert label is the truth. What counts as a 'commitment or action' is a judgement call, and I have no inter-annotator figure for this dataset. 19 of 320 test paragraphs are ones where all five model runs agree with each other and disagree with the experts (12 misses, 7 false alarms). If those are label errors the model is better than 63.0%; if they are a real definition gap it is not. I have not adjudicated them, and doing so with the same model would be circular.

What a sceptic attacks first

Transfer. Paragraphs are already pre-filtered as climate-related, so this is not a transition-plan document test: finding the paragraphs inside a 200-page plan is a prior step this module does not measure. The label is binary and paragraph-level. It is not extraction of a target (what, how much, by when), which has no public labelled set here. Paragraphs of one company share style, and the data has no company id, so the resample treats them as independent; the interval is narrower than it should be by an amount I cannot measure here.

Not claimed

Nothing here is forward-looking, so there is no backtest; the held-out test set plays that role. Model aliases move: the resolved IDs are in results/eval.json and the cached replies in results/llm_raw.jsonl. The strict-versus-terse prompt trade-off is shown on one data set and one model family.

Next

Label a sample of real transition-plan PDFs myself, under a written codebook with a second pass to measure my own consistency, extract target, year and scope, and score the same way.